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RESEARCH ARTICLE

# Improving Perception of Usability Through User Interface Design Patterns to Optimize Information Architecture for Cognitive Benefits and User Satisfaction in Massive Open Online Courses

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**ABSTRACT** This study explores the impact of user interface design patterns on usability, cognitive load, and user satisfaction for Massive Open Online Courses using small-screen devices. An empirical approach was adopted, involving 232 university students who voluntarily participated in the experiment. Prototypes of three well-known Massive Open Online Courses platforms (i.e., Coursera, Udemy, and edX) were developed to assess how various user interface design patterns influence user experience. The findings revealed that the aesthetic design of Coursera, including color scheme, content organization, was perceived as the most visually appealing, while Udemy received higher ratings for its typography, i.e, font size, type, and button shape. Coursera also outperformed the other platforms in terms of navigation (e.g., tab navigation, hamburger menu, drop-down, floating action button, listview), customization features (e.g., search filters, font, and background settings), and feedback mechanisms (e.g., toast messages, error alerts, progress indicators, confirmation prompts, and system status updates). Overall, participants reported higher satisfaction with Coursera, and its interface was associated with a lower cognitive load compared to Udemy and edX. These results underscore the importance of thoughtful user interface design in enhancing usability and reducing cognitive effort in mobile learning applications.

**INDEX TERMS** Cognitive load, design patterns, information architecture, massive open online courses, usability, user interfaces, user-satisfaction.

## I. INTRODUCTION

the significance of eLearning technologies has grown substantially [1]. Among these technologies, Massive Open Online Courses (MOOCs) offering platforms have become one of the most popular trends in online education, helping students to achieve their intended goals [2], [3]. The global adoption of digital education platforms has accelerated the need to evaluate their User Interface (UI) design for a better user experience.

User satisfaction and learning outcomes in these platforms are significantly influenced by UI design quality [1], [4]. Due to poor UI design, MOOCs face significant adoption challenges due to poor usability and User interface (UI) design problems [4], [5]. UI design strongly impacts mental resources load, and poor UI design leads to cognitive

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burden [5], [6], [7]. This overwhelming cognitive burden can escalate into frustration, irritation, and boredom, ultimately leading to system abandonment [8]. Furthermore, research indicates that fewer than 10% of MOOC participants complete their courses [4], [5].

Previous research has largely concentrated on examining user experiences through fully functional desktop web browsers to explore design implications [9]. Website interface on desktop utilizes precise, mouse-driven interactions, smartphones rely on touch-based interfaces, where the human finger, being less accurate than a mouse, often results in increased error rates and reduced interaction precision [10], [11]. Desktop users typically engage through indirect manipulation, while mobile users interact via direct touch, demanding a different approach to interface design [6], [11].

Design principles from Human Computer Interaction (HCI), including visibility, affordance, feedback, and consistency, are foundational to effective UI design [2], [12], [13]. Moreover, the diversity in user needs across web and mobile platforms introduces added complexity in UI design [13], [14], [15]. The design of handheld devices (e.g, smartphones, tablets) has introduced new UI design patterns, including menu layouts and interaction techniques. The precise use of these UI patterns to fit for heightened interaction is still challenging for UI designers. This may be due to the limited screen size, tiny controls, unregulated navigation, lack of organization, inappropriate layout, poor responsiveness, and poor legibility [6], [16].

Usability is one of the key metrics to evaluate UI design and cognitive load [6], [12], [17]. A well-designed UI enhances ease of use and overall user experience [18]. Usability assesses whether the system is efficient, effective, safe, easy to learn, remember, and use while also ensuring user satisfaction [19]. A well-designed and aesthetically attractive interface, characterized by harmonious color schemes, uniform layouts, and visually pleasing elements, significantly influences the user's perceptions of a system's usability and encourages their engagement [18], [20], [21]. More importantly, for smaller screen controls, such as menu icons, navigation through menus, and moving across tabs, impacts usability [7], [22].

Several studies have emphasized that poor usability poses significant challenges to MOOC adoption [1], [14]. Research on the usability of MOOCs is growing by focusing on challenges related to small screens, limited input methods, and changing user contexts [12], [23]. Therefore, further studies are required to propose suitable UI design patterns to minimize the usability issues.

Besides usability, cognitive engagement is another key metric for evaluating UI design. Cognitive engagement refers to cognitive benefits, which are essential for assessing the success of e-learning platforms [15], [24]. Cognitive load refers to the mental effort required by users to execute tasks on mobile devices [25], and it is an important factor in determining the user experiences [26]. Suleri et al. [27] observed that the UI design strategies based on UI patterns significantly lower the users' cognitive workload compared to traditional design strategies.

UI patterns for smaller screens influence cognitive functions, and poor UI design consumes substantial mental resources and potentially causes cognitive load [6], [7]. On smaller screens, users face challenges during different tasks such as typing and searching, ultimately affecting learning performance [6], [12], [17], [22], [28]. Although some recent investigations have explored the impact of specific UI design patterns on MOOC usability, most have focused primarily on navigational elements [14], overlooking other critical design components that may also significantly influence user engagement, satisfaction, and overall learning outcomes.

To overcome cognitive load and usability challenges, effective UI design strategies are necessary. E-learning service providers are eager to create effective interface designs for online education, aiming to enhance student engagement and their continued use [13].

Despite the growing emphasis on user experience in mobile-based learning environments, the relationship between UI design patterns and their impact on usability, cognitive load, and user satisfaction remains underexplored in the context of MOOCs. This study aims to investigate how common UI design patterns influence user interaction by improving usability and minimizing cognitive burden, ultimately enhancing user satisfaction. To address this gap, an empirical research methodology is adopted through university students as participants. The primary objective is to examine the effect of UI design patterns on improving usability and satisfaction in online learning environments by reducing cognitive load. The findings will be used to provide practical guidelines for the development community, along with well-suited design UI patterns and interaction genres for online learning environments. Accordingly, an experimental prototype of three well-known MOOC platforms (i.e, Coursera, edX, and Udemy) will be simulated, and the participants will be required to interact with the employed experimental prototypes. Their experiences will be assessed using well-established tools, such as the CEACAM usability model [29], the System Usability Scale (SUS) [14], the cognitive load (NASA Task Load Index (NASA-TLX)), and user satisfaction.

To accomplish the above-mentioned objectives, the following questions are defined.

### A. RESEARCH QUESTIONS

**RQ1**: Which specific feature of UI design patterns influences perceived usability?

**RQ2**: Which employed MOOCs (i.e., Coursera, Udemy, and edX) UI induces the least cognitive load?

**RQ3**: Which employed MOOCs (i.e., Coursera, Udemy, and edX) UI is more satisfying for the user's needs?

The remainder of this paper is organized as follows: Section II reviews the related work; Section III outlines the research methodology; Section IV presents the experimental analysis and discussion; Section V reports the results; Section VI highlights the implications of the findings; and Section VII concludes the paper with a summary of key insights, limitations, and directions for future work.

## II. RELATED WORK

Digitalization has changed the world. Every age group uses mobile technology according to their requirements [30]. Smartphones are ubiquitous, efficient, and capable of instant communication, and create an immersive environment for knowledge sharing [31]. Online learning has become a common practice for both students and teachers in acquiring and sharing knowledge and technological advancements with increased accessibility. In addition, educational platforms have adopted MOOCs [32]. MOOCs are innovative eLearning platforms that deliver educational services on a large scale, free from location and time limitations [13], [33].

Despite the widespread adoption of MOOCs, enhancing user experience and fostering continued usage remains a key focus in MOOC research [13]. MOOC platforms rely heavily on user interface (UI) design, which plays a crucial role in shaping learners' performance and overall experiences. A well-crafted UI evokes positive emotions, enhances user satisfaction, and significantly contributes to improved learning outcomes by creating a seamless and engaging learning experience. [6], [34], [35], [36], [37], [38]. The core principles of UI design: visibility, affordance, feedback, and consistency, are fundamental to human-computer interaction. Visibility ensures critical functions are easily seen, enhancing task efficiency. Affordance uses visual cues to suggest functionality, reducing task time. Feedback provides immediate system responses, minimizing user uncertainty. Consistency maintains uniform patterns, lowering cognitive load and learning effort. These principles collectively optimize user-system interaction when effectively applied, though their implementation varies across platforms [2], [12], [13], [39]. In contrast, neglecting the principles of user interface design often leads to ineffective user interfaces [9], [40]. Previous research [1], [7], [13], [30], [41], [42], [43], [44] has emphasized that inadequate usability and high cognitive load due to poor UI design represent significant obstacles to MOOC adoption, as suboptimal designs adversely impact the eLearning experience. Mobile learning is generally less convenient than desktop learning, with one of the primary challenges being the limited screen size and tiny controls [28].

Website interfaces that utilize precise mouse-driven interactions, smartphones rely on touch-based interfaces, where the human finger is less accurate than a mouse, often results in increased error rates and reduced interaction precision [6], [11], [45]. Furthermore, mobile devices must present the same content in a more compact space, which can further hinder usability and increase cognitive load [46]. Another critical consideration is the significant variation in user interaction patterns between large-screen desktop interfaces and mobile interfaces [11], [46]. While website users typically engage through indirect manipulation, mobile users interact via direct touch, demanding a different approach to interface design. These limitations necessitate the identification and mitigation of mobile-specific usability issues, as they directly influence user satisfaction, engagement, and continued use [37], [47].

Rapid advancements in smartphone user interfaces have led to new UI design patterns, including menu layouts and interaction techniques [12], [23]. UI design patterns have developed into a valuable resource for documenting and reusing proven solutions. The structure of a design pattern must include specific parameters to make it useful. The same design pattern can be represented differently to visualize more extensive or simplified structures [48]. Researchers have compiled comprehensive catalogs of UI mobile patterns, creating a rich repository of implementable examples [49]. An interface that effectively incorporates UI design patterns enables users to achieve their goals more efficiently compared to one that lacks such support [27], [50].

A well-crafted navigation can enhance system usability and positively influence users' motivational states [51]. Moreover, color not only helps the users during navigational and functional tasks but also during the whole interaction time [8]. Various researchers in the elegant literature [52], [53] discussed the impact of color on usability, emotions, attitude, and intention to purchase. Furthermore, positioning buttons and controls in appropriate areas of mobile applications enables users to access necessary information more easily and efficiently [54]. In another study, Faisal et al. [5] emphasized that enhancing cognitive engagement requires attention to specific design quality factors such as font style (e.g., sans-serif), font size (14–22 px), appropriate color combinations (e.g., blue-white, grey-black, blue), layout structure, spacing, and element grouping based on Gestalt principles.

In HCI, usability has become an essential part of the system development process [32]. Usability is an important part of the user experience [55]. It plays a crucial role in the effectiveness of mobile learning applications, as well-designed UI significantly influences the user's perceptions of a system's usability and encourages their engagement [18], [20], [21]. In another study, Nikou [36] argued that effective user interface design improves usability and increases user capacity and knowledge. Smaller screen sizes and use of UI design patterns majorly influence usability and alter individuals' usage behaviour and adaptation [7], [56].

The content on the interface that is poorly organized, insufficient, or inconsistent in quality leads to significant usability issues [5]. Unexperienced users face even more usability issues while interacting with mobile devices, such as identifying menu icons, navigating through menus, and moving between tabs [22]. Tiny controls alongside small screens can reduce usability due to ergonomic issues, which include a crowded interface, unregulated navigation, lack of organization, inappropriate layout, poor responsiveness, inadequate legibility, and limited space [6], [16]. Users who are not experts have usability issues while interacting with mobile devices, such as identifying menu icons, navigating through menus, and moving between tabs [22]. Studies indicate that users, especially older adults, often experience more challenges when navigating complex mobile interfaces, which can lead to frustration and decreased satisfaction [14]. Satisfaction is characterized as freedom from discomfort and positive attitudes toward using the product [57]. Usability evaluation serves as a crucial method for assessing user satisfaction. This satisfaction is fundamental to usability, with Nielsen noting that it is a key factor that can directly influence usability levels [58].

Besides usability, cognitive engagement is essential in evaluating the success and intention to use e-learning platforms, as it significantly influences information processing and continued usage [24]. Cognitive load, by definition, refers to the mental effort required by users to execute tasks on mobile devices [25]. The growing use of mobile devices intensifies competition for users' attention, making cognitive workload a key factor in the user experience [15]. The use of MOOCs on small-screen devices influences cognitive load, recent findings show that poor interface design further increases cognitive load and negatively impacts learning performance [6], [12], [17]. A small screen can make typing and searching more difficult, increasing cognitive load and potentially hinder learning performance [22], [28]. For instance, research exploring student interactions with learning management systems found that intuitive navigation and clear visual hierarchies correlate strongly with increased engagement and reduced cognitive load [59]. Navigation and clear visual hierarchies in learning management systems strongly correlate with increased engagement and reduced cognitive load [59]. The NASA Task Load Index (NASA-TLX) is a commonly utilized tool to evaluate cognitive load. It measures six factors: mental demand (MD), physical demand (PD), temporal demand (TD), performance, effort (PE), and frustration (F) [60]. Ali et al. [61] applied NASA-TLX to the usability testing of users of different age groups. Gani et al. [62] argued that the NASA Task Load Index (NASA TLX) is an effective instrument for collecting subjective feedback from participants.

User behavior is the most important factor in user-centered UI design [67]. eLearning service providers are striving to create inclusive interface designs for online education to retain students and continue online learning [13]. Utilizing well-designed eLearning technologies to access educational content can foster an online learning environment that is more accessible and engaging [6]. Current mobile UI design trends emphasize the creation of more intuitive and engaging user experiences [68]. However, it is worth noting that most research describes a limited number of designs and provides minimal information regarding user usability difficulties and the need for more research in capturing and describing mobile UI design patterns [55]. The following Table 1 summarizes prior studies related to user interface design patterns and principles, as discussed in previous literature. To summarize, existing research has shown that usability and cognitive load are key factors for MOOCs adoption and retention. However, there is a need to investigate how they impact MOOCs' learning performance.

To the best of our knowledge, in this respect no experimental research has been performed. In addition, the impact of UI patterns on usability and cognitive load, and how the information architecture influences satisfaction, has not been studied in existing research. To address this research gap, this study proposes an empirical research-based approach to determine the relationship between UI design patterns and their effects on usability, cognitive load, and satisfaction.

## III. METHODOLOGY

This study aims to explore the role of UI design patterns in improving the perception of usability by minimizing the cognitive load of MOOCs to achieve overall satisfaction. The study primarily adopted an empirical approach by collecting data using a questionnaire survey. The data was collected from university students who volunteered to participate in the study. Initially, experimental prototypes were developed for well-known MOOC platforms(i.e., Coursera, Udemy, and edX). The developed prototypes use different UI design patterns based on the standard guidelines and were similar to the employed MOOCs. The categorization of the employed UI patterns was primarily based on the industrial standards and scientific studies, such as hamburger menu, breadcrumb navigation, and tabbed navigation patterns [49], which were used to help the users move through, or navigate back and forth between the different sections. Typography, color scheme, and background color patterns used for aesthetics [58], [69]. Search and filter patterns are used for customized search [49], [58]. The content display card pattern is used to show the course outlines, progress indicators, and course details [67], [70]. Cardview patterns are used to adjust the content according to screen size [71]. Notification and feedback patterns are used for short pop-up messages that indicate task completion, errors, or updates [49]. The list menu pattern is used to vertically display the courses on the screen [49]. The video player interface pattern is used, which includes all controls related to videos. Moreover, a major color scheme is used in Coursera (white, blue, light and dark grey, orange), Udemy (white, purple, light and dark grey, blue), and edX(white, greenish black, red, light and dark grey). The sans-serif font style is used in all three MOOCs (see Fig. 2, Fig. 3, Fig. 4).

An experimental procedure was designed to conduct a study to assess the user experiences for these developed platforms. After developing prototypes, the users were asked to access these developed platforms. Participants could register, enroll in courses, and download lecture notes and videos using the prototypes. Quantitative data was analyzed

**TABLE 1.** Design Related Studies.

| Design Principles/Patterns | Relationship Bahaviour/Attitude | Application Type (Context of Use) | References |
|---|---|---|---|
| UI Design (Appearance, Organization and Information Architecture) | Continued intension to use | e-Learning | [6] |
| Navigation patterns | Usability | Social media | [14] |
| Visual Design, Navigation Design, and Responsiveness | Continued intension to use | e-Learning | [5] |
| User Control, Responsiveness, Navigation, and Visual Design | Trust and intension to use | e-Learning | [7] |
| Animation, Color,and Font | Emotions | e-commerce | [54] |
| Animation, images,and layout | Usability | Healthcare | [63] |
| Visual Design | Trust | e-commerce | [45] |
| Visual Design Principles | Engagement and User Experience | e-Learning | [64] |
| Visual Design | Reliability and usability | e-Learning | [65] |
| Navigation | Usability | e-Learning | [66] |
| UI Design | Cognitive load | e-Learning | [12] |
| Visual Design and Navigation | Emotions | Commercial | [54] |
| Visual Design | Continued intension to use | Social media | [37] |
| Visual Design | Cognitive load | e-Learning | [23] |
| Visual Design | Cognitive load and Usability | Commercial | [61] |

using SPSS 29.0. Both descriptive and inferential (e.g., paired t-test) statistical techniques were used to perform the analysis. Initially, it was important to check the reliability of the questionnaire; therefore, to assess the reliability and internal consistency, Cronbach's Alpha was computed. This study is centered around three objectives, which are divided into three studies: study one assessed the impact of UI patterns on perceived usability, study two explored which employed MOOC induces the least cognitive load on user experiences, and the third study examined which employed MOOC was more satisfying for the participants.

### *A. PARTICIPANTS AND DATA COLLECTION*

This study used a survey-based approach to collect data from university students. The population consisted of undergraduate students enrolled in a university. Undergraduate students from the domain of computer science were selected as participants in this experiment, comprising 60% males and 40% females. Before the experiment, all students were briefed on the tasks they were required to perform. University students were considered the most suitable sample for this study due to their frequent use of the eLearning environment. Before the actual experimentation, required permissions were applied for and obtained from the IRB board of concerned departments and institutions. At the start of the academic session, the prototypes of the selected MOOCs were introduced to the participants in collaboration with university management and faculty. Participants were encouraged to use the learning platform (MOOCs) for enrollment and to complete course-related activities on their mobile devices. The relevant and sufficient guidance was provided to the participants during the various sessions. Participants were asked to use each prototype for two months, and at the end of each session, participants were asked to share their experiences and perceptions of each prototype by using a survey including questions related to perceived usability, cognitive load, and satisfaction. A survey scale was prepared by using Google Forms and integrated within the experimental prototypes. Table 2 shows the demographic description of participants, and Fig. 1 shows the Linear Process Diagram.

**TABLE 2.** Sample Data (Participant Details).

| N | Educational Background | Gender % | Age | Device Type |
|---|---|---|---|---|
| 232 | Undergraduate | Male 60%<br>Female 40% | 17-23 | Android / iPhone |

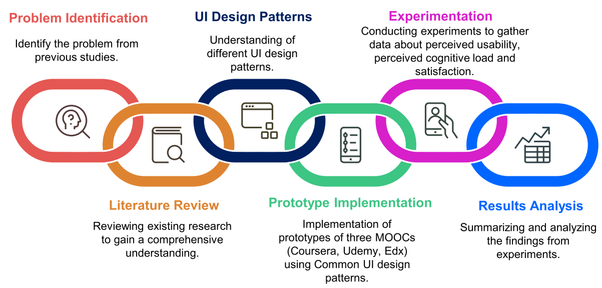


**FIGURE 1.** Linear process diagram.

## IV. EXPERIMENTAL ANALYSIS AND DISCUSSION

Participants were engaged with the prototypes and completed their tasks, which included registration, enrollment, downloading lecture notes and videos, engaging in discussions with their peers through the conversation portal courses, navigating within the prototypes, and exploring the visual design, typography, font quality, interaction design, customization, feedback, and user controls.

### *A. UI PATTERNS AND PERCEIVED USABILITY (RQ1)*

This study aims to evaluate the influence of UI design patterns on perceived usability. To assess the perceived usability of the interface design, the survey scale is taken primarily from a recent study [29]. The statistical results showed the significant differences between the prototypes, providing insight into how the UI design influences perceived usability. The results offer a comprehensive understanding of user preferences and the effectiveness of different design

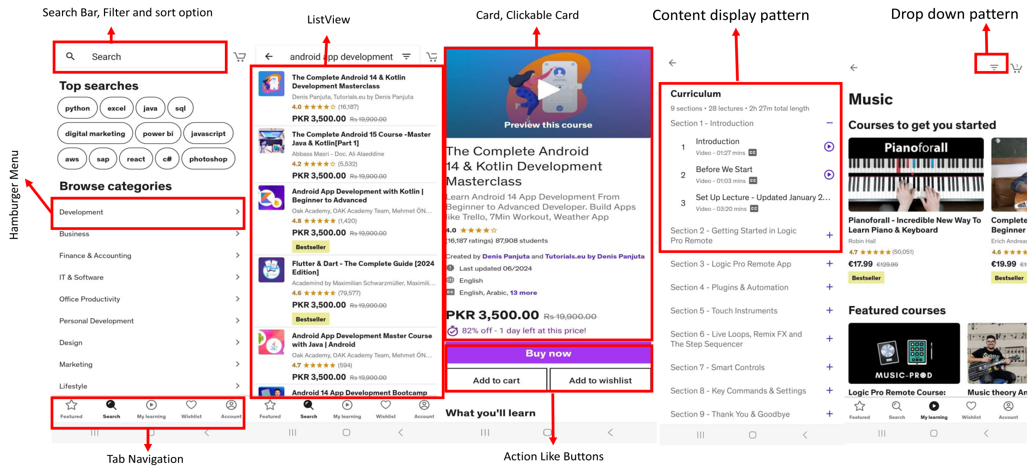


**FIGURE 2.** MOOC of Udemy.

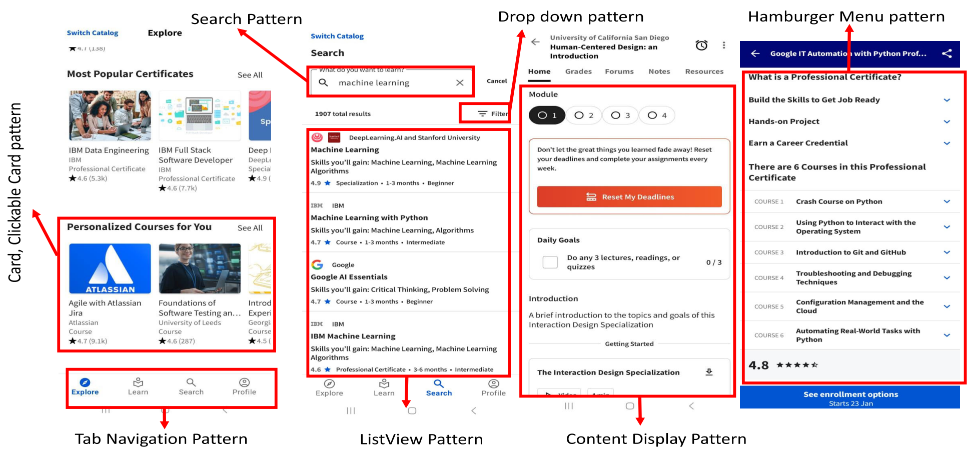


**FIGURE 3.** MOOC of Coursera.

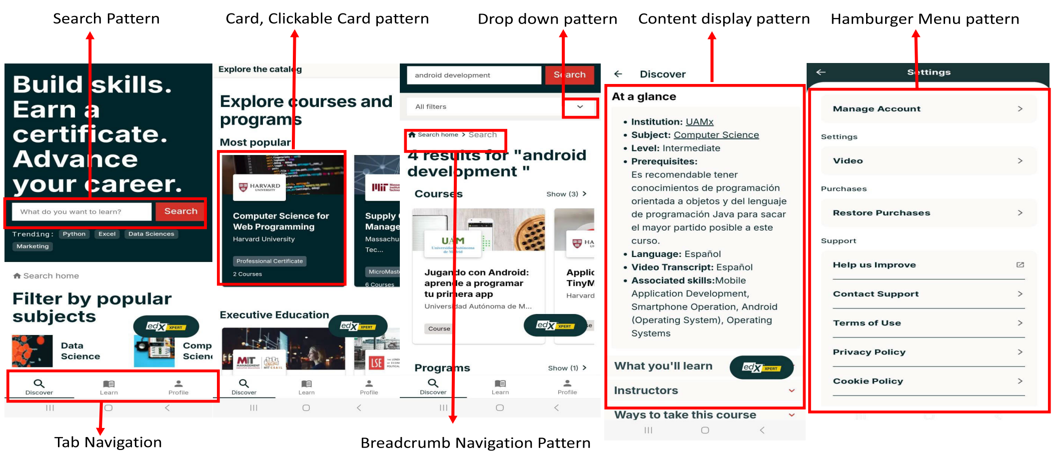


**FIGURE 4.** MOOC of edX.

**TABLE 3. Perceived Usability Questionnaire.**

| UI Design | ID | Items | Cronbach's ($\alpha$) |
|---|---|---|---|
| Aesthetics | A1 | The color scheme used to design the interface is appealing and attractive to the student. | 0.776 |
| | A2 | The information presented adjusts to the screen size. | |
| Typography | T1 | The font type, size, and spaces allow easy reading of information. | |
| | T2 | The interface design is consistent in style, font size, buttons, colors, etc.(same design through the whole application) | |
| Navigation | N1 | The desired contents or basic tasks are easily accessible from the main page in three or less clicks. | 0.750 |
| | N2 | The application clearly presents the search option to help students find content. | |
| | N3 | The application clearly presents an option that allows us to return to the main menu. | |
| | N4 | There are instructions of use for certain functions in applications, these are easy to find and understand (help option) | |
| | N5 | The dimension and proximity of the tactile button or selection controls are appropriate to be easily selected with fingers. | |
| Customization | C1 | The application allows changing the font size and type. | 0.830 |
| | C2 | The application allows changing the screen background and font color (e.g., Clear text over a dark background and vice versa). | |
| | C3 | The application provides options for advanced configuration and is easy to find. | |
| | C4 | The application UI design adjusts itself to the different screen sizes. (New Item) | |
| | C5 | The application allows users to choose among different languages. | |
| | C6 | The application allows you to choose different ways of input and output. | |
| | C7 | The application allows you to convert text to speech or transcribe text from a voice dictation. | |
| | C8 | The application allows basic functions to customize user preferences(search, setting). (New Item) | |
| Responsiveness | R1 | The application provides precise and instant feedback related to actions performed by the user. | 0.840 |
| | R2 | The application provides precise feedback related to the system state (ex., A status bar) showing information about the progress on an action. | |
| | R3 | The application provides information about the units you have already mastered and what you have yet to complete or study. | |
| | R4 | The application provides reminders related to the goals established by the student. | |
| | R5 | During the self-evaluation, if the student makes a mistake, the app provides opportunities for correction. | |
| | R6 | If the student makes a mistake during self-evaluations, the app will provide explanations about the correct responses. | |
| | R7 | The application provides information when actions are performed that could generate non-desired effects (ex., warnings or confirmations to prevent mistakes). | |

**TABLE 4. Paired t-test analysis of Visual Design/ Aesthetics.**

| ID# | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| A1 | Coursera | vs | edX | **2.353** | 0.019 |
| A1 | Coursera | vs | Udemy | **2.010** | 0.046 |
| A1 | Udemy | vs | edX | -0.566 | 0.572 |
| A2 | Coursera | vs | edX | 0.670 | 0.504 |
| A2 | Coursera | vs | Udemy | 0.336 | 0.737 |
| A2 | Udemy | vs | edX | -0.285 | 0.776 |

**TABLE 5. Paired t-test analysis of Visual Design/ Typography.**

| ID | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| T1 | Coursera | vs | edX | **2.027** | 0.044 |
| T1 | Coursera | vs | Udemy | 0.278 | 0.781 |
| T1 | edX | vs | Udemy | **-2.038** | 0.043 |
| T2 | Coursera | vs | edX | **2.725** | 0.007 |
| T2 | Coursera | vs | Udemy | 0.313 | 0.755 |
| T2 | edX | vs | Udemy | **-2.091** | 0.038 |

approaches. Different aspects of usability were considered during the study experiment, which are discussed in this section.

### 1) VISUAL DESIGN

Visual design plays a prominent role by enhancing readability and recognition, and drawing users' attention to important information. Furthermore, key elements such as precise organization, symmetry, grouping, order, animation, color, font, and layout create an attractive and visually pleasing appearance [12], [26], [27], [72]. In this research, visual design is discussed in two further categories. The reliability of the questions under visual design was computed, which was acceptable with the value of $\alpha = 0.776$ (see Table 3).

***Aesthetics*** refers to the characteristics of a stimulus, including the appeal and attractiveness of an interface, conveyed through graphics, color, and animation [73]. Students responded to two questions (A1, A2) in the category of aesthetics, which were about the color scheme and the content organization of MOOCs. The result of A1 shows a statistically significant difference was found between Coursera and edX (t = 2.353 and p = 0.019) and between Coursera and Udemy (t = 2.010 and p = 0.046). The results of the A2 did not give any significant differences (see Table 4). The results indicate that the UI patterns, i.e, color scheme, content organization, and clickable card patterns, used in Coursera's aesthetics are more appealing compared to the others (see Table 6).

In the same context, Faisal et al. [5] argued that visual arrangement refers to the planned organization and placement of elements within a user interface to heighten usability, readability, and overall user experience. Aesthetics is recognized as a fundamental element of perceived usability [20]. Furthermore, Faisal et al. [6] also examined the influence of aesthetics on involvement dimensions, including cognitive and affective involvement, ultimately driving the intention to continue using the system.

***Typography*** refers to the visual appeal, attractiveness, and readability of the text in the UI, designed to capture and engage the user's attention. Font quality describes the visual presentation and arrangement of text, enhancing readability and supporting effective information processing [6]. Two questions(T1, T2) were posed concerning font style, color, size, and button color, and the reliability of these questions was considered acceptable (see Table 3). Furthermore, the statistical result of T1 indicates a significant difference in typography between Coursera and edX (t = 2.027 and p = 0.044). However, no significant difference was found between Coursera and Udemy. An important difference was observed between edX and Udemy(t = −2.338 and p = 0.043). Similarly, the result of the T2 shows a significant difference between Coursera and edX (t = 2.725 and p = 0.007), and no difference was found between Coursera and

**TABLE 6. Descriptive Statistics of Visual Design.**

| UI Design | Prototype | ID | Mean | Median | Mode | SD |
|---|---|---|---|---|---|---|
| Aesthetics | Coursera | A1 | **4.228** | 4.000 | 4.00 | 0.769 |
| | | A2 | 4.181 | 4.000 | 4.00 | 0.779 |
| Typography | | T1 | 4.245 | 4.000 | 4.00 | 0.620 |
| | | T2 | 4.189 | 4.000 | 4.00 | 0.737 |
| Aesthetics | edX | A1 | 4.034 | 4.000 | 4.00 | 0.992 |
| | | A2 | 4.133 | 4.000 | 4.00 | 0.780 |
| Typography | | T1 | 4.112 | 4.000 | 4.00 | 0.787 |
| | | T2 | 4.000 | 4.000 | 4.00 | 0.789 |
| Aesthetics | Udemy | A1 | 4.081 | 4.000 | 4.00 | 0.760 |
| | | A2 | 4.155 | 4.000 | 4.00 | 0.848 |
| Typography | | T1 | **4.262** | 4.000 | 4.00 | 0.693 |
| | | T2 | **4.168** | 4.000 | 4.00 | 0.796 |

**TABLE 7. Pair t-test analysis of Navigation.**

| ID | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| N1 | Coursera | vs | edX | **2.005** | 0.046 |
| N1 | Coursera | vs | Udemy | **2.714** | 0.007 |
| N1 | edx | vs | Udemy | 0.923 | 0.357 |
| N2 | Coursera | vs | edX | 1.454 | 0.147 |
| N2 | Coursera | vs | Udemy | -0.118 | 0.906 |
| N2 | edx | vs | Udemy | -1.516 | 0.131 |
| N3 | Coursera | vs | edX | 0.000 | 1.000 |
| N3 | Coursera | vs | Udemy | -0.115 | 0.909 |
| N3 | edX | vs | Udemy | -0.122 | 0.903 |
| N4 | Coursera | vs | edX | 0.481 | 0.631 |
| N4 | Coursera | vs | Udemy | 0.123 | 0.902 |
| N4 | edX | vs | Udemy | -0.319 | 0.750 |
| N5 | Coursera | vs | edX | 0.925 | 0.356 |
| N5 | Coursera | vs | Udemy | **2.233** | 0.027 |
| N5 | edX | vs | Udemy | 1.602 | 0.110 |

Udemy. A significant difference was observed between edX and Udemy($t = -2.091$ and $p = 0.038$). The results are shown in Table 5.

The analysis of mean values indicated that users rated UI patterns of Udemy's typography ( i.e, font size, font type, button color) more positively, with a score of mean = 4.262, compared to Coursera and edX (see Table 6).

In a recent study, Vecino et al [74] argued that typography significantly influences usability. Similarly, in other studies [75], [76], the authors highlighted the significant role of typography in improving user experience, trust, and satisfaction. Faisal et al. [8] also argued that typographical features influence usability. Well-structured spacing between lines and words, appropriate font color, style, and readable font size create a visually appealing and reliable interface [52].

### 2) NAVIGATION

Navigation design defines how users access different parts of the UI, impacting satisfaction and usability. Having well-structured navigation reduces frustration and improves the user experience. Designers use it to guide users efficiently through tasks and content [5], [14], [77].

In the navigation category, five questions were presented to evaluate the participants' perceptions of the navigation experience. The reliability of the questionnaire was 0.750 (see Table 3). The result of the first question (N1), which was about the easy accessibility of the content, indicated a significant difference between Coursera and edX ( $t = 2.005$ and $p = 0.046$), and between Coursera and Udemy($t = 2.714$ and $p = 0.007$). However, no significant difference was found between edX and Udemy. However, the results of N2, N3, and N4 indicate that there were no significant differences between the three prototypes in response to these two questions. Another significant difference was found in N5 between Coursera and Udemy ($t = 2.233$ and $p = 0.027$). The result of the last question (N5), which was about user controls, indicates a significant difference between Coursera and Udemy($t = 2.2333$ and $p = 0.027$). However, there was no significant difference between Coursera and edX. Similarly, there was no significant difference between edX and Udemy (see Table 7). However, after observing the mean values as illustrated in Table 8, participants perceived Coursera's

**TABLE 8. Descriptive Statistics of Navigation.**

| UI Design | Prototype | ID | Mean | Median | Mode | SD |
|---|---|---|---|---|---|---|
| Navigation | Coursera | N1 | **4.241** | 4.000 | 4.00 | 0.697 |
| | | N2 | 4.202 | 4.000 | 4.00 | 0.771 |
| | | N3 | 4.081 | 4.000 | 4.00 | 0.841 |
| | | N4 | 4.073 | 4.000 | 4.00 | 0.689 |
| | | N5 | **4.168** | 4.000 | 4.00 | 0.779 |
| | edX | N1 | 4.103 | 4.000 | 4.00 | 0.777 |
| | | N2 | 4.094 | 4.000 | 4.00 | 0.832 |
| | | N3 | 4.081 | 4.000 | 4.00 | 0.809 |
| | | N4 | 4.038 | 4.000 | 4.00 | 0.874 |
| | | N5 | 4.103 | 4.000 | 4.00 | 0.725 |
| | Udemy | N1 | 4.030 | 4.000 | 4.00 | 0.927 |
| | | N2 | 4.211 | 4.000 | 4.00 | 0.780 |
| | | N3 | 4.090 | 4.000 | 4.00 | 0.730 |
| | | N4 | 4.064 | 4.000 | 4.00 | 0.816 |
| | | N5 | 3.961 | 4.000 | 4.00 | 1.045 |

UI patterns (i.e., tab navigation pattern, hamburger menu pattern, drop-down pattern, floating action button pattern, listview pattern) to provide better navigation than the other two prototypes.

Navigation patterns improve usability and satisfaction [14]. Incorporated suitable mobile navigation elements such as patterns, transitions, hierarchy, buttons, links, paths, and schemes, as these factors are important to enhance user experience [54]. Similarly, Oh et al. [43] identified navigation as a crucial design element of the MOOC platform that significantly impacts learning experiences. Another recent study by Faisal et al. [5] argued that user control features are essential in learning environments, as they empower users to manage their interactions.

### 3) INTERACTION DESIGN (CUSTOMIZATION)

Interaction design refers to how users manage and engage with the information presented, encompassing aspects such as customization and content management [78]. It also refers to collaborative tools within online learning systems that enhance interaction [66]. Customization refers to the adaptation of content to align with individual needs. This involves the ease with which portals can be modified

**TABLE 9.** Pair t-test analysis of Customization.

| ID | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| C1 | Coursera | vs | edX | -0.993 | 0.322 |
| C1 | Coursera | vs | Udemy | -1.270 | 0.205 |
| C1 | edX | vs | Udemy | -0.273 | 0.785 |
| C2 | Coursera | vs | edX | -0.737 | 0.462 |
| C2 | Coursera | vs | Udemy | 0.202 | 0.840 |
| C2 | edX | vs | Udemy | 0.988 | 0.324 |
| C3 | Coursera | vs | edX | 0.277 | 0.782 |
| C3 | Coursera | vs | Udemy | 0.616 | 0.538 |
| C3 | edX | vs | Udemy | 0.370 | 0.712 |
| C4 | Coursera | vs | edX | **2.096** | 0.037 |
| C4 | Coursera | vs | Udemy | 0.320 | 0.749 |
| C4 | edX | vs | Udemy | -1.954 | 0.052 |
| C5 | Coursera | vs | edX | **2.532** | 0.012 |
| C5 | Coursera | vs | Udemy | 1.618 | 0.107 |
| C5 | edX | vs | Udemy | -0.596 | 0.552 |
| C6 | Coursera | vs | edX | 0.497 | 0.620 |
| C6 | Coursera | vs | Udemy | 0.437 | 0.663 |
| C6 | edX | vs | Udemy | -0.044 | 0.965 |
| C7 | Coursera | vs | edX | -0.944 | 0.346 |
| C7 | Coursera | vs | Udemy | -1.844 | 0.066 |
| C7 | edX | vs | Udemy | -0.944 | 0.346 |
| C8 | Coursera | vs | edX | **2.248** | 0.026 |
| C8 | Coursera | vs | Udemy | 0.636 | 0.526 |
| C8 | edX | vs | Udemy | -1.461 | 0.146 |

**TABLE 10.** Descriptive Statistics of Customization.

| UI Design | Prototype | ID | Mean | Median | Mode | SD |
|---|---|---|---|---|---|---|
| Customization | Coursera | C1 | 3.719 | 4.000 | 4.00 | 1.082 |
| | | C2 | 3.870 | 4.000 | 4.00 | 1.032 |
| | | C3 | 4.047 | 4.000 | 4.00 | 0.879 |
| | | C4 | **4.232** | 4.000 | 4.00 | 0.871 |
| | | C5 | **4.163** | 4.000 | 4.00 | 0.788 |
| | | C6 | 3.939 | 4.000 | 4.00 | 0.923 |
| | | C7 | 3.702 | 4.000 | 4.00 | 1.113 |
| | | C8 | **4.034** | 4.000 | 4.00 | 0.906 |
| | edX | C1 | 3.810 | 4.000 | 4.00 | 1.052 |
| | | C2 | 3.939 | 4.000 | 4.00 | 0.956 |
| | | C3 | 4.025 | 4.000 | 4.00 | 0.872 |
| | | C4 | 4.047 | 4.000 | 4.00 | 1.024 |
| | | C5 | 3.965 | 4.000 | 4.00 | 0.952 |
| | | C6 | 3.896 | 4.000 | 4.00 | 1.018 |
| | | C7 | 3.793 | 4.000 | 4.00 | 1.089 |
| | | C8 | 3.836 | 4.000 | 4.00 | 1.064 |
| | Udemy | C1 | 3.836 | 4.000 | 4.00 | 1.023 |
| | | C2 | 3.853 | 4.00 | 4.00 | 0.942 |
| | | C3 | 3.995 | 4.000 | 4.00 | 0.880 |
| | | C4 | 4.206 | 4.000 | 4.00 | 0.816 |
| | | C5 | 4.021 | 4.000 | 4.00 | 1.074 |
| | | C6 | 3.900 | 4.000 | 4.00 | 1.094 |
| | | C7 | 3.883 | 4.000 | 4.00 | 0.993 |
| | | C8 | 3.978 | 4.000 | 4.00 | 0986 |

or customized according to user preferences, which is recognized as a critical quality factor in improving customer experiences and influencing utilization behavior [79], [80].

In this category, participants were asked eight questions (C1 to C8) related to UI customization. The reliability of the questions was also acceptable, with the value of ( *alpha* = 0.830), see Table 3. The analysis indicated a statistically significant difference between Coursera and edX for question C4, which addressed adoptability ($t = 2.096$, $p = 0.037$). Similarly, a significant difference was observed for question C5, concerning the availability of options in different languages ($t = 2.523$, $p = 0.012$). Additionally, question C8, related to customization options, also revealed a significant difference between the two platforms ($t = 2.248$, $p = 0.026$), see Table 9. The participants preferred Coursera's responsive UI patterns (i.e., filter and sort pattern, adaptive layout patterns, and responsive cardview patterns, and tabbed navigation pattern for settings) for its better customization, and Udemy is the second most preferred among the three prototypes(see Table 10).

In a recent study, Minas et al. [81] also explained that in interaction design, customization is an important factor in the user interface that increases usability and improves the user experience. Zhan et al. [82] also discovered that customization potentially increases user engagement.

#### 4) INTERACTION DESIGN (RESPONSIVENESS)

Responsiveness is an important factor in user interface usability that makes an application easy to use. An effective design should continuously inform users about the system's state by providing timely and appropriate feedback [83]. Feedback includes progress, encouragement, support, precise alerts, system status, and error prevention [29].

Seven questions (R1 to R7) were designed to evaluate participants' perceptions about responsiveness across the three prototypes. The reliability analysis confirmed an acceptable consistency level, with a Cronbach's alpha value of $\alpha = 0.840$ (see Table 3). The statistical analysis for R3, concerning course completion progress information, revealed a significant difference between Coursera and edX ($t = 2.149$, $p = 0.033$). Likewise, R5, which addressed error correction support, showed a statistically significant difference between the same platforms ($t = 2.042$, $p = 0.042$). Additionally, the result for R6, related to corrective feedback with explanations, indicated a significant difference between edX and Udemy ($t = -2.829$, $p = 0.005$). No further significant differences were identified (see Table 11). The findings suggest that participants perceived Coursera's UI patterns for feedback (i.e., toast message patterns, alert dialogue pattern, inline validation pattern) as the most effective in terms of interaction responsiveness (see Table 12).

Fath et al. [84] also examined that the system's usability and responsiveness correlate positively. In a recent study, Faisal et al. [5] established that responsiveness significantly influences focused immersion, temporal dissociation, and curiosity of users. Previously, Multiple studies [33], [50], [85] have identified responsiveness as a key factor influencing trust, satisfaction, engagement, cognitive aspects, and the intention to continue using a system.

### B. PERCEIVED COGNITIVE LOAD (RQ2)

This study seeks to examine which MOOC platforms' UI design information architecture, which is a combination of different UI design patterns, contributes to lower perceived

**TABLE 11.** Pair t-test analysis of Responsiveness.

| ID | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| R1 | Coursera | vs | edX | 0.791 | 0.430 |
| R1 | Coursera | vs | Udemy | 0.000 | 1.000 |
| R1 | edX | vs | Udemy | -0.735 | 0.463 |
| R2 | Coursera | vs | edX | 0.964 | 0.336 |
| R2 | Coursera | vs | Udemy | 0.578 | 0.564 |
| R2 | edX | vs | Udemy | -0.391 | 0.696 |
| R3 | Coursera | vs | edX | **2.149** | 0.033 |
| R3 | Coursera | vs | Udemy | 1.467 | 0.144 |
| R3 | edX | vs | Udemy | -0.621 | 0.535 |
| R4 | Coursera | vs | edX | 1.326 | 0.186 |
| R4 | Coursera | vs | Udemy | 0.737 | 0.462 |
| R4 | edX | vs | Udemy | -0.613 | 0.540 |
| R5 | Coursera | vs | edX | **2.042** | 0.042 |
| R5 | Coursera | vs | Udemy | 0.273 | 0.785 |
| R5 | edX | vs | Udemy | -1.889 | 0.060 |
| R6 | Coursera | vs | edX | 1.305 | 0.193 |
| R6 | Coursera | vs | Udemy | -1.821 | 0.070 |
| R6 | edX | vs | Udemy | **-2.829** | 0.005 |
| R7 | Coursera | vs | edX | 0.820 | 0.413 |
| R7 | Coursera | vs | Udemy | -0.754 | 0.452 |
| R7 | edX | vs | Udemy | -1.578 | 0.116 |

**TABLE 12.** Descriptive Statistics of Responsiveness.

| UI Design | Prototype | ID | Mean | Median | Mode | SD |
|---|---|---|---|---|---|---|
| Responsiveness | Coursera | R1 | 4.172 | 4.000 | 4.00 | 0.735 |
| | | R2 | 4.125 | 4.000 | 4.00 | 0.771 |
| | | R3 | **4.150** | 4.000 | 4.00 | 0.836 |
| | | R4 | 4.077 | 4.000 | 4.00 | 0.927 |
| | | R5 | **4.017** | 4.000 | 4.00 | 0.934 |
| | | R6 | **4.126** | 4.000 | 4.00 | 0.939 |
| | | R7 | 3.956 | 4.000 | 4.00 | 0.957 |
| | edX | R1 | 4.120 | 4.000 | 4.00 | 0.746 |
| | | R2 | 4.051 | 4.000 | 4.00 | 0.861 |
| | | R3 | 3.978 | 4.000 | 4.00 | 0.909 |
| | | R4 | 3.965 | 4.000 | 4.00 | 0.957 |
| | | R5 | 3.814 | 4.000 | 4.00 | 1.122 |
| | | R6 | 3.801 | 4.000 | 4.00 | 1.094 |
| | | R7 | 3.883 | 4.000 | 4.00 | 0.997 |
| | Udemy | R1 | 4.172 | 4.000 | 4.00 | 0.759 |
| | | R2 | 4.081 | 4.00 | 4.00 | 0.804 |
| | | R3 | 4.030 | 4.000 | 4.00 | 0.879 |
| | | R4 | 4.017 | 4.000 | 4.00 | 0.842 |
| | | R5 | 3.955 | 4.000 | 4.00 | 0.904 |
| | | R6 | 4.073 | 4.000 | 4.00 | 0.862 |
| | | R7 | 4.021 | 4.000 | 4.00 | 0.865 |

cognitive load on the user experience. The NASA Task Load Index (NASA-TLX) is a commonly utilized tool to evaluate cognitive load. It measures six factors: mental demand (MD), physical demand (PD), temporal demand (TD), performance, effort (PE), and frustration (F) [60]. Ali et al. [61] applied NASA-TLX to the usability testing of users of different age groups. Gani et al. [62] argued that the NASA Task Load Index (NASA TLX) is an effective instrument for collecting subjective feedback from participants. The measurement scale utilized for the NASA-TLX in this study ranges from 1 to 5, with 1 representing the lowest level and 5 indicating the highest. Participants were asked five questions regarding three prototypes to assess the impact of the information architecture of UI design on cognitive load. The specific NASA-TLX questions are presented in Table 13.

The participants shared their insights, responding to six NASA-TLX questions for each prototype. The quantitative data was analyzed using SPSS 29.0, and the reliability of questions was also calculated, which is acceptable with the value of $\alpha = 0.711$. A paired t-test was used to determine whether there were significant differences in cognitive load across the three prototypes.

#### 1) MENTAL DEMAND

Mental demand refers to the level of cognitive and perceptual effort required to complete a task, including activities such as thinking, deciding, calculating, remembering, and searching [15]. Participants were asked to evaluate how easy or difficult, straightforward or complex, and precise the tasks were on each MOOC platform. The analysis revealed significant differences in perceived mental demand across all prototypes (see Table 14). Participants reported the highest level of mental demand while using Udemy ($M = 3.975$), whereas edX was associated with a moderately lower level ($M = 3.617$). Coursera was perceived as the least mentally demanding, with a mean score of 3.13 (see Table 15, Fig. 5).

#### 2) PHYSICAL DEMAND

Physical demand refers to the extent to which users felt pressured by the pace of tasks, the amount of physical effort required, and whether the task completion speed was perceived as fast or slow [86]. The analysis of participant responses revealed statistically significant differences in perceived physical demand across the three MOOC platforms (see Table 16). Udemy was reported to have the highest physical demand ($M = 4.024$), followed by edX ($M = 3.786$). In contrast, Coursera was perceived as significantly less physically demanding ( $M = 30243$) (see Table 17, Fig. 5).

#### 3) TEMPORAL DEMAND

Temporal demand refers to the time pressure and pacing of tasks, which can influence a user's stress levels—whether the pace is perceived as comfortable and manageable or rushed and overwhelming [87]. The analysis revealed a statistically significant difference between Coursera and edX ($t = 2.494$ and $p = 0.013$) and between Udemy and edX ($t = -2.725$ and $p = 0.007$). However, no significant difference was found between Coursera and Udemy (see Table 18). Although differences in mean values across the three platforms were relatively small, Udemy was associated with the highest temporal demand ($M = 3.358$), followed by edX ($M = 3.311$). Coursera was perceived as having the lowest temporal demand ($M = 3.034$) (see Table 19, Fig. 5).

#### 4) PERFORMANCE

Performance refers to participants' perceptions of their success in accomplishing task goals and their overall satisfaction with their performance [87]. As shown in Table 20, a statistically significant difference was observed between

**TABLE 13.** NASA-TLX Questionnaire.

| Q# | NASA-TLX Questionnaire | | Cronbach's alpha($\alpha$) |
|---|---|---|---|
| 1 | Mental Demand | How mentally demanding was the task? | 0.711 |
| 2 | Physical Demand | How physically demanding was the task? | |
| 3 | Temporal Demand | How hurried or rushed was the pace of the task? | |
| 4 | Performance | How successful were you in accomplishing what you were asked to do? | |
| 5 | Effort | How hard did you have to work to accomplish your level of performance? | |
| 6 | Frustration | How insecure, discouraged, irritated, stressed, and annoyed were you? | |

**TABLE 14.** Pair t-test analysis of Mental Demand.

| ID | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| MD | Coursera | vs | edX | **4.537** | 0.000 |
| MD | Coursera | vs | Udemy | **-6.923** | 0.000 |
| MD | edX | vs | Udemy | **-3.103** | 0.002 |

**TABLE 15.** Descriptive Statistics of Mental Demand.

| Q# | Prototype | ID | Mean | Median | Mode | SD |
|---|---|---|---|---|---|---|
| Mental Demand | Coursera | MD | **3.134** | 4.000 | 4.00 | 1.275 |
| | edX | MD | 3.621 | 3.000 | 3.00 | 1.018 |
| | Udemy | MD | 3.975 | 4.000 | 5.00 | 1.119 |

**TABLE 16.** Pair t-test analysis of Physical Demand.

| ID | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| PD | Coursera | vs | edX | **5.103** | 0.000 |
| PD | Coursera | vs | Udemy | **-7.347** | 0.000 |
| PD | edX | vs | Udemy | **-2.088** | 0.038 |

**TABLE 17.** Descriptive Statistics of Physical Demand.

| Q# | Prototype | ID | Mean | Median | Mode | SD |
|---|---|---|---|---|---|---|
| Physical Demand | Coursera | PD | **3.243** | 3.000 | 3.00 | 1.198 |
| | edX | PD | 3.786 | 4.000 | 4.00 | 0.969 |
| | Udemy | PD | 4.024 | 4.000 | 5.00 | 1.055 |

**TABLE 18.** Pair t-test analysis of Temporal Demand.

| ID | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| TD | Coursera | vs | edX | **2.494** | 0.013 |
| TD | Coursera | vs | Udemy | **-2.725** | 0.007 |
| TD | edX | vs | Udemy | -0.218 | 0.828 |

**TABLE 19.** Descriptive Statistics of Temporal Demand.

| Q# | Prototype | ID | Mean | Median | Mode | SD |
|---|---|---|---|---|---|---|
| Temporal Demand | Coursera | TD | **3.034** | 3.000 | 3.00 | 1.212 |
| | edX | TD | 3.311 | 4.000 | 4.00 | 1.137 |
| | Udemy | TD | 3.358 | 4.000 | 5.00 | 1.212 |

**TABLE 20.** Pair t-test analysis of Performance.

| ID | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| P | Coursera | vs | edX | **-6.057** | 0.000 |
| P | Coursera | vs | Udemy | 6.278 | 0.000 |
| P | edX | vs | Udemy | **0.681** | 0.496 |

**TABLE 21.** Descriptive Statistics of Performance.

| Q# | Prototype | ID | Mean | Median | Mode | SD |
|---|---|---|---|---|---|---|
| Performance | Coursera | P | **2.955** | 3.000 | 2.00 | 1.217 |
| | edX | P | 2.283 | 2.000 | 2.00 | 0.997 |
| | Udemy | P | 2.204 | 2.000 | 2.00 | 1.184 |

**TABLE 22.** Pair t-test analysis of Effort.

| ID | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| E | Coursera | vs | edX | **4.862** | 0.000 |
| E | Coursera | vs | Udemy | **-6.338** | 0.000 |
| E | edX | vs | Udemy | -1.018 | 0.310 |

Coursera and edX ($t = -6.057$ and $p = 0.000$), as well as between Coursera and Udemy ($t = 6.278$ and $p = 0.000$). No significant difference was found between Udemy and edX. The findings indicate that Coursera was rated highest in terms of perceived performance ($M = 2.955$), followed by edX, with Udemy ranked lowest (see Table 21, Fig. 5).

### 5) EFFORT

Effort refers to the mental effort participants had to exert to achieve their desired level of performance while using the mobile application. Consider the cognitive load required to navigate the interface, complete tasks, and process information. Was the UI intuitive and efficient, or did it require significant effort to understand and use effectively [87]. The researchers identified a significant difference between Coursera and edX($t = 4.862$ and $p = 0.000$), as well as a notable difference between Coursera and Udemy($t = -6.338$ and $p = 0.000$). However, no significant difference was observed between edX and Udemy. The results are displayed in the Table 22. Participants rated Udemy the highest, indicating it required the most effort to complete tasks compared to the other two prototypes. edX was ranked second, while Coursera was perceived as the least demanding in terms of effort( see Table 23, Fig. 5).

### 6) FRUSTRATION

The Frustration Level assesses how users felt during the task, whether they experienced insecurity, discouragement,

**TABLE 23.** Descriptive Statistics of Effort.

| Q# | Prototype | ID | Mean | Median | Mode | SD |
|---|---|---|---|---|---|---|
| Effort | Coursera | E | **2.885** | 3.000 | 3.00 | 1.154 |
| | edX | E | 3.447 | 4.000 | 4.00 | 1.023 |
| | Udemy | E | 3.611 | 4.000 | 4.00 | 1.186 |

**TABLE 24.** Pair t-test analysis of Frustration.

| ID | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| F | Coursera | vs | edX | **7.541** | 0.000 |
| F | Coursera | vs | Udemy | **-9.192** | 0.000 |
| F | edX | vs | Udemy | -1.576 | 0.117 |

**TABLE 25.** Descriptive Statistics of Frustration.

| Q# | Prototype | ID | Mean | Median | Mode | SD |
|---|---|---|---|---|---|---|
| Frustration | Coursera | F | **2.950** | 3.000 | 2.00 | 1.325 |
| | edX | F | 3.835 | 4.000 | 4.00 | 1.099 |
| | Udemy | F | 4.059 | 4.000 | 5.00 | 1.125 |

irritation, stress, and annoyance, or instead felt secure, gratified, content, relaxed, and complacent [87]. The results in Table 24 indicate a significant difference between Coursera and edX($t = 7.514$ and $p = 0.000$), as well as between Coursera and Udemy($t = -9.192$ and $p = 0.000$). However, no significant difference was observed between edX and Udemy. The participants felt more frustrated while using Udemy, and edX was in second place. Participants perceived that they felt less frustrated while using Coursera (see Table 25, Fig. 5).

The user interface design influences the cognitive load [12]. Younas et al. [7] argued that ineffective interface design can elevate cognitive load, ultimately diminishing the learner's intention to use the system. Alasmari [88] also explored the effect of mobiles with small screen sizes on cognitive load in the learning environment.

### C. SATISFACTION (RQ3)

Satisfaction is how comfortable the user is with the design [89]. Consequently, this study aimed to evaluate participants' overall satisfaction with the prototypes used in the experimentation. The usability encompasses satisfaction as a key component [90]. User satisfaction represents the users' overall perception and attitude toward the experience of using mobile applications [91]. Several factors contribute to overall satisfaction, including entertainment, usefulness, aesthetic appeal, and a sense of reward [55].

To check the participant's overall satisfaction while using these prototypes, two questions (S1, S2) associated with satisfaction were asked from this category, and the reliability of these questions was Cronbach's alpha value of $\alpha = 0.666$ (see Table 26). Furthermore, according to Table 27, the analysis indicates that the researchers found a significant difference between Coursera and edX, with a t-sig value

**TABLE 26.** Satisfaction Questionnaire.

| | ID | Items | Cronbach's alpha ($\alpha$) |
|---|---|---|---|
| Satisfaction | S1 | Overall, I am satisfied with how easy it is to use this system | 0.666 |
| | S2 | If I had access to the mobile app over the coming months, I believe that I would use it instead of other systems | |

**TABLE 27.** Pair t-test analysis of Satisfaction.

| ID | Prototypes | | | t-value | Sig (two-side p) |
|---|---|---|---|---|---|
| S1 | Coursera | vs | edX | 2.151 | 0.033 |
| S1 | Coursera | vs | Udemy | 0.606 | 0.545 |
| S1 | edX | vs | Udemy | -1.464 | 0.145 |
| S2 | Coursera | vs | edX | 2.648 | 0.009 |
| S2 | Coursera | vs | Udemy | 2.192 | 0.029 |
| S2 | edX | vs | Udemy | -0.503 | 0.615 |

**TABLE 28.** Descriptive Statistics of Satisfaction.

| | Prototype | ID | Mean | Median | Mode | SD |
|---|---|---|---|---|---|---|
| Satisfaction | Coursera | S1 | 4.194 | 4.000 | 4.00 | 0.735 |
| | Coursera | S2 | 4.228 | 4.000 | 4.00 | 0.957 |
| | edX | S1 | 4.043 | 4.000 | 4.00 | 0.746 |
| | edX | S2 | 4.000 | 4.000 | 4.00 | 0.997 |
| | Udemy | S1 | 4.150 | 4.000 | 4.00 | 0.759 |
| | Udemy | S2 | 4.043 | 4.000 | 4.00 | 0.865 |

of 2.151 ($p = 0.033$) in S1. In S2, a significant difference was found between Coursera and edX with a t(sig) value of 2.648(0.009), and another significant difference was found between Coursera and Udemy with a t(sig) value of 2.192($p = 0.029$). The mean value indicates that participants feel more satisfied using the Coursera prototype (see Table 28 and Fig. 6).

## V. RESULTS

The study, conducted in three phases, evaluated the impact of UI design patterns on perceived usability, cognitive load, and user satisfaction across three MOOC platforms: Coursera, Udemy, and edX. Our empirical analysis of three MOOC platforms reveals significant variations in UI pattern effectiveness across usability, cognitive load, and satisfaction metrics. In terms of perceived usability, Coursera was rated highest for its aesthetically appealing elements such as color schemes, content organization, and clickable card patterns, while Udemy received more favorable feedback on typography. Coursera also outperformed the others in navigation due to effective use of tab navigation, hamburger menus, and floating action buttons. For customization and interaction responsiveness, Coursera was again preferred, followed by Udemy. In assessing cognitive load, participants reported that Udemy imposed the highest mental, physical, and temporal demands, while Coursera was perceived as the least demanding and most efficient, with higher performance ratings and lower levels of required effort and frustration. Finally, in terms of overall user satisfaction, the participants

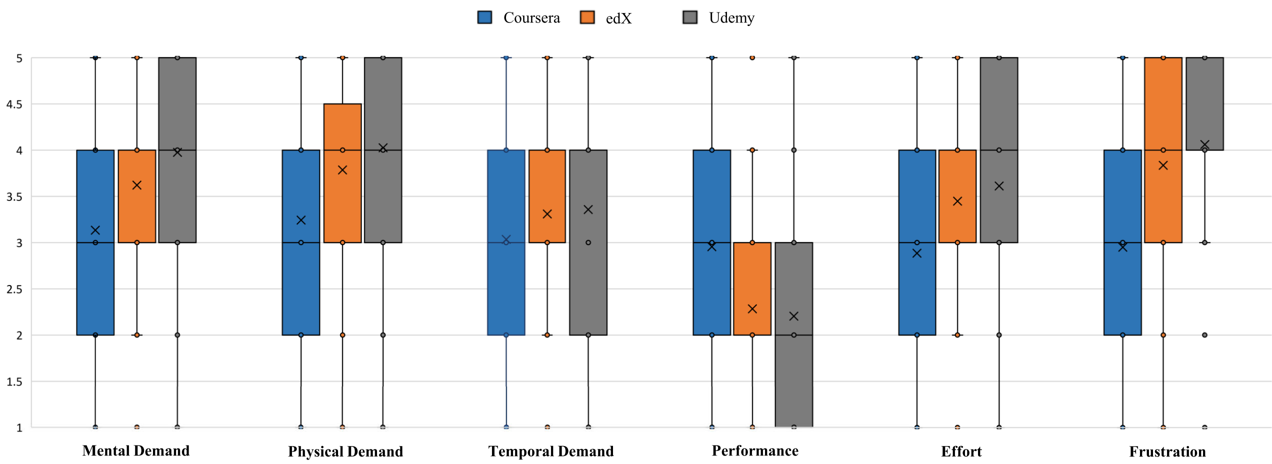


**FIGURE 5.** Box Plot of NASA-TLX Cognitive Load Scores Across Coursera, edX, and Udemy.

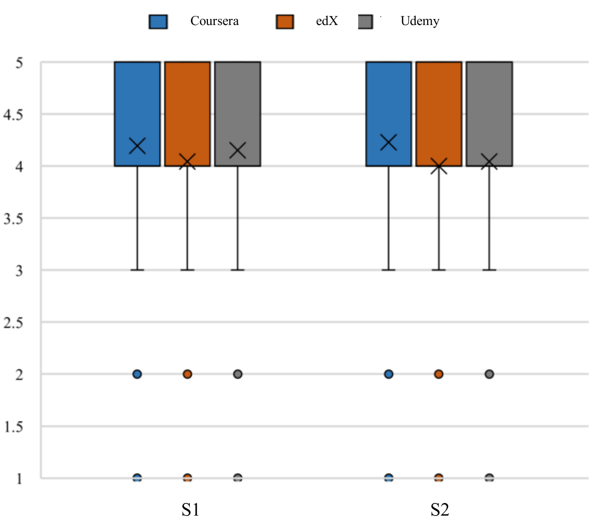


**FIGURE 6.** Box Plot of User Satisfaction Ratings Across MOOCs.

expressed the greatest comfort and satisfaction with the Coursera prototype, further affirming its superior UI design in all the dimensions evaluated.

## VI. IMPLICATION

The findings of this study carry several important implications for UI/UX designers, developers, and researchers working in the domain of mobile learning platforms. First, the study underscores the critical role of UI design patterns in shaping user perceptions of usability, cognitive load, and satisfaction. The superior performance of Coursera's UI prototype across all three evaluation phases suggests that strategic implementation of specific patterns such as visually cohesive aesthetics, intuitive navigation, responsive feedback, and customizable elements, can substantially improve the learning experience on mobile devices. Designers should prioritize such patterns to create interfaces that are not only functionally effective but also emotionally engaging and cognitively efficient.

For developers, these results emphasize the importance of integrating adaptable and responsive UI elements that accommodate diverse user needs and device constraints. The ability of Coursera's prototype to minimize user effort and frustration demonstrates how thoughtful UI design can reduce barriers to engagement and task completion, particularly in educational contexts where cognitive overload may hinder learning.

From a research perspective, this study contributes to the growing body of knowledge linking micro-level UI patterns to macro-level user experience outcomes. It moves beyond generic assessments of usability by offering a pattern-level analysis, thus providing a more granular understanding of what design features matter most. Future studies can build upon this work by exploring how these patterns perform in different cultural, demographic, or domain-specific contexts, and how they interact with individual differences in learning style or cognitive capacity.

Overall, the study reinforces that user-centered UI design is not merely an aesthetic concern but a foundational component of effective digital learning systems. These implications encourage interdisciplinary collaboration between educators, designers, and engineers to create mobile interfaces that are accessible, efficient, and satisfying for a wide range of learners.

## VII. CONCLUSION, LIMITATIONS, AND FUTURE WORK

This study provides empirical evidence on the role of UI design patterns in shaping user experience across mobile-based MOOC platforms, with a focus on perceived usability, cognitive load, and user satisfaction. Through a three-phase evaluation of prototypes modeled on Coursera, Udemy, and edX, the findings highlight Coursera's UI design patterns as the most effective in enhancing user experience across all assessed dimensions. The study confirms that specific UI components such as color schemes, navigation mechanisms, customization features, and responsive feedback patterns substantially influence perceived usability. Typography elements were also found to be critical, particularly in Udemy's prototype, suggesting that legibility and visual hierarchy play a key role in user engagement.

The cognitive load analysis further indicated that Coursera's design minimized mental, physical, and temporal demand, leading to higher perceived performance and lower frustration. In terms of satisfaction, users consistently favored Coursera's interface, emphasizing the value of cohesive and intuitive information architecture in learning environments. These findings offer both theoretical and practical contributions: for researchers, they advance understanding of the interplay between UI patterns and user experience constructs; for practitioners, they inform evidence-based UI design strategies for mobile educational platforms.

Despite the contributions, this study has a few limitations. First, the use of experimental prototypes, while controlled for consistency, may not fully replicate real-world usage contexts or content dynamics of live MOOC platforms. Second, the participant pool may lack diversity in demographics such as age, cultural background, or technological familiarity, which could influence user perceptions. Third, the study focused solely on three MOOC platforms, which limits the generalizability of the findings across the broader e-learning ecosystem.

Future work could explore a more diverse range of users to assess how demographic factors such as gender, age, or educational background influence interaction with UI patterns. Additionally, future research might investigate the role of device-specific variables, such as screen size and operating system, in shaping usability and cognitive responses. Longitudinal studies could also offer insight into how user preferences evolve with continued use, providing a deeper understanding of sustained engagement and satisfaction in mobile learning environments.

**Data availability**: The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

**Conflict of interest**: The authors declare that there are no conflicts of interest regarding the publication of this paper.

**Consent to participate:** The research involves the participation of the National Textile University and the FAST National University, Chiniot-Faisalabad campus. The corresponding authors are free to contact any of the people involved in the study to seek further clarification and information.

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## REFERENCES

[1] M. Altalhi, "Toward the sustainability of mobile learning applications in higher education: An empirical study," *Universal Access Inf. Soc.*, vol. 23, no. 3, pp. 1103–1113, Aug. 2024.

[2] M. Soegaard, "Interaction design foundation: The basic of user experience design. SL: Interaction design foundation," Interaction-Design.Org, Tech. Rep., pp. 29–33, 2002.

[3] G. R. Morales, A. P. Gallegos, P. V. Torres-Carrión, and S. C. Carrión, "Comparative evaluation of the overall user experience of two MOOC platforms: Coursera and OpenCampus," in *Developments and Advances in Defense and Security: Proceedings of MICRADS 2022*. Cham, Switzerland: Springer, 2023, pp. 179–190.

[4] P. Rodríguez-Torrico, S. San-Martín, and R. San José-Cabezudo, "What drives M-shoppers to continue using mobile devices to buy?" *J. Marketing Theory Pract.*, vol. 27, no. 1, pp. 83–102, Jan. 2019.

[5] C. M. N. Faisal, A. Younas, J. De Andrés, D. Fernandez-Lanvin, and M. Gonzalez-Rodriguez, "The effects of appearance and organization and information architecture on the continued intention to use MOOCs," *Universal Access Inf. Soc.*, vol. 24, no. 3, pp. 2219–2237, Aug. 2025.

[6] C. M. N. Faisal, D. Fernandez-Lanvin, J. De Andrés, and M. Gonzalez-Rodriguez, "Design quality in building behavioral intention through affective and cognitive involvement for e-learning on smartphones," *Internet Res.*, vol. 30, no. 6, pp. 1631–1663, Jul. 2020.

[7] A. Younas, C. M. N. Faisal, M. A. Habib, R. Ashraf, and M. Ahmad, "Role of design attributes to determine the intention to use online learning via cognitive beliefs," *IEEE Access*, vol. 9, pp. 94181–94202, 2021.

[8] C. M. N. Faisal, J. De Andres-Suarez, M. Gonzalez-Rodriguez, D. Fernandez-Lanvin, M. Ahmad, and M. A. Habib, "Impact of web design features on irritation for e-commerce websites," in *Proc. 33rd Annu. ACM Symp. Appl. Comput.* New York, NY, USA: ACM Press, Apr. 2018, pp. 656–663.

[9] D. Cyr, M. Head, E. Lim, and A. Stibe, "Using the elaboration likelihood model to examine online persuasion through website design," *Inf. Manage.*, vol. 55, no. 7, pp. 807–821, Nov. 2018.

[10] J. M. N. Ribeiro, "Web design patterns for mobile devices," Fac. Fine Arts, Univ. Porto, Portoo, Portugal, Tech. Rep., 2012. [Online]. Available: https://repositorio-aberto.up.pt/handle/10216/64860?mode=full

[11] H. Hoehle, R. Aljafari, and V. Venkatesh, "Leveraging Microsoft's mobile usability guidelines: Conceptualizing and developing scales for mobile application usability," *Int. J. Hum.-Comput. Stud.*, vol. 89, pp. 35–53, May 2016.

[12] M. A. Faudzi, Z. C. Cob, M. Ghazali, R. Omar, and S. A. Sharudin, "User interface design in mobile learning applications: Developing and evaluating a questionnaire for measuring learners' extraneous cognitive load," *Heliyon*, vol. 10, no. 18, Sep. 2024, Art. no. e37494.

[13] D. Tao, P. Fu, Y. Wang, T. Zhang, and X. Qu, "Key characteristics in designing massive open online courses (MOOCs) for user acceptance: An application of the extended technology acceptance model," *Interact. Learn. Environments*, vol. 30, no. 5, pp. 882–895, May 2022.

[14] M. Umar, I. Hussain, T. Mahmood, H. T. Mirza, and C. M. N. Faisal, "Design strategies to minimize mobile usability issues in navigation design patterns," *Information*, vol. 15, no. 11, p. 732, Nov. 2024.

[15] T. Kosch, J. Karolus, J. Zagermann, H. Reiterer, A. Schmidt, and P. W. Woźniak, "A survey on measuring cognitive workload in human–computer interaction," *ACM Comput. Surveys*, vol. 55, no. 13s, pp. 1–39, Dec. 2023.

[16] T. Natarajan, S. A. Balasubramanian, and D. L. Kasilingam, "The moderating role of device type and age of users on the intention to use mobile shopping applications," *Technol. Soc.*, vol. 53, pp. 79–90, May 2018.

[17] J. Dunaway and S. Soroka, "Smartphone-size screens constrain cognitive access to video news stories," *Inf., Commun. Soc.*, vol. 24, no. 1, pp. 69–84, Jan. 2021.

[18] M. Xiang, L. Sha, and Z. Qilin, "User satisfaction model of functional animation design in the digital technology perspective based on TAM," in *Proc. Int. Conf. Innov. Design Digit. Technol. (ICIDDT)*, Dec. 2020, pp. 301–305.

[19] M. Ballantyne, A. Jha, A. Jacobsen, J. S. Hawker, and Y. N. El-Glaly, "Study of accessibility guidelines of mobile applications," in *Proc. 17th Int. Conf. Mobile Ubiquitous Multimedia*, Nov. 2018, pp. 305–315.

[20] M. Soui, M. Chouchane, N. Bessghaier, M. W. Mkaouer, and M. Kessentini, ''On the impact of aesthetic defects on the maintainability of mobile graphical user interfaces: An empirical study,'' *Inf. Syst. Frontiers*, vol. 24, no. 2, pp. 659–676, Apr. 2022.

[21] V. Sharma, K. Jangir, M. Gupta, and R. Rupeika-Apoga, ''Does service quality matter in FinTech payment services? An integrated SERVQUAL and TAM approach,'' *Int. J. Inf. Manage. Data Insights*, vol. 4, no. 2, Nov. 2024, Art. no. 100252.

[22] A. H. Cano Bejar, S. Ray, and Y. H. Huang, ''Fighting for the status quo: Threat to tech self-esteem and opposition to competing smartphones,'' *Inf. Manage.*, vol. 60, no. 2, Mar. 2023, Art. no. 103748.

[23] B. Curum and K. K. Khedo, ''Cognitive load management in mobile learning systems: Principles and theories,'' *J. Comput. Educ.*, vol. 8, no. 1, pp. 109–136, Mar. 2021.

[24] A. Occa and S. E. Morgan, ''The role of cognitive absorption in the persuasiveness of multimedia messages,'' *Comput. Educ.*, vol. 176, Jan. 2022, Art. no. 104363.

[25] P. Weichbroth, ''Usability of mobile applications: A systematic literature study,'' *IEEE Access*, vol. 8, pp. 55563–55577, 2020.

[26] N. H. Nasrudin, S. Ahmad, K. A. Salleh, and L. A. Azahari, ''A systematic review of user mental models on applications sustainability,'' *Int. J. Sustain. Construct. Eng. Technol.*, vol. 14, no. 3, pp. 376–389, 2023.

[27] S. Suleri, Y. Hajimiri, and M. Jarke, ''Impact of using UI design patterns on the workload of rapid prototyping of smartphone applications: An experimental study,'' in *Proc. 22nd Int. Conf. Hum.-Comput. Interact. Mobile Devices Services*, Oct. 2020, pp. 1–5.

[28] I. D. Sabukunze and A. Arakaza, ''User experience analysis on mobile application design using user experience questionnaire,'' *Indonesian J. Inf. Syst.*, pp. 15–26, Aug. 2021.

[29] C. X. Navarro-Cota, A. I. Molina, M. A. Redondo, and C. Lacave, ''A comprehensive usability measurement tool for m-learning applications,'' *IEEE Trans. Educ.*, vol. 67, no. 2, pp. 209–223, Apr. 2024.

[30] S. J. Barnes, A. D. Pressey, and E. Scornavacca, ''Mobile ubiquity: Understanding the relationship between cognitive absorption, smartphone addiction and social network services,'' *Comput. Hum. Behav.*, vol. 90, pp. 246–258, Jan. 2019.

[31] N. Tuli and A. Mantri, ''Evaluating usability of mobile-based augmented reality learning environments for early childhood,'' *Int. J. Hum.-Comput. Interact.*, vol. 37, no. 9, pp. 815–827, May 2021.

[32] F. Hanis, M. Asri, D. Singh, Z. Mansor, and H. Norman, ''A review of cross-cultural design to improve user engagement for learning management system,'' *KSII Trans. Internet Inf. Syst.*, vol. 18, no. 2, pp. 397–419, 2024.

[33] (2024). *The Big Three Platforms Revisited: The Latest on Coursera, EDX, & Udemy*. [Online]. Available: https://www.encoura.org/resources/wake-up-call/the-big-three-platforms-revisited-the-latest-on-coursera-edx-udemy/

[34] K. Ghoulam, B. Bouikhalene, A. Babori, and N. Falih, ''Exploring the impact of mobile devices in electronics E-learning: A case study evaluating the effectiveness of mobile learning applications in the field of electronics and sensors,'' *Adv. Mobile Learn. Educ. Res.*, vol. 4, no. 2, pp. 1058–1072, Jun. 2024.

[35] Z. Shao and K. Chen, ''Understanding individuals' engagement and continuance intention of MOOCs: The effect of interactivity and the role of gender,'' *Internet Res.*, vol. 31, no. 4, pp. 1262–1289, Jul. 2021.

[36] S. A. Nikou, ''Factors influencing student teachers' intention to use mobile augmented reality in primary science teaching,'' *Educ. Inf. Technol.*, vol. 29, pp. 15353–15374, Jan. 2024.

[37] A. Tarute, S. Nikou, and R. Gatautis, ''Mobile application driven consumer engagement,'' *Telematics Informat.*, vol. 34, no. 4, pp. 145–156, Jul. 2017.

[38] S. A. Nikou and A. A. Economides, ''Mobile-based assessment: Investigating the factors that influence behavioral intention to use,'' *Comput. Educ.*, vol. 109, pp. 56–73, Jun. 2017.

[39] R. Alnanih and O. Ormandjieva, ''Mapping HCI principles to design quality of mobile user interfaces in healthcare applications,'' *Proc. Comput. Sci.*, vol. 94, pp. 75–82, Jun. 2016.

[40] J. Vanderdonckt, ''Automatic user interface generation from declarative model,'' in *Proc. 2nd Int. Conf. Comput.-Aided Design User Interfaces*, 2024, pp. 4–61. [Online]. Available: https://books.google.com.pk/books?

[41] A. Liapis, V. Maratou, T. Panagiotakopoulos, C. Katsanos, and A. Kameas, ''UX evaluation of open MOOC platforms: A comparative study between moodle and open edX combining user interaction metrics and wearable biosensors,'' *Interact. Learn. Environ.*, vol. 31, no. 10, pp. 6841–6855, Dec. 2023.

[42] Y.-M. Cheng, ''Can gamification and interface design aesthetics lead to MOOCs' success?'' *Edu. + Training*, vol. 63, no. 9, pp. 1346–1375, Nov. 2021.

[43] E. Oh, M. Cho, and Y. Chang, ''Learners' perspectives on MOOC design,'' *Distance Educ.*, vol. 44, no. 3, pp. 476–494, 2023.

[44] P.-A. Cinquin, P. Guitton, and H. Sauzéon, ''Designing accessible MOOCs to expand educational opportunities for persons with cognitive impairments,'' *Behav. Inf. Technol.*, vol. 40, no. 11, pp. 1101–1119, Aug. 2021.

[45] D. Cyr, ''Website design, trust and culture: An eight country investigation,'' *Electron. Commerce Res. Appl.*, vol. 12, no. 6, pp. 373–385, Nov. 2013.

[46] P. Lugtig and V. Toepoel, ''The use of PCs, smartphones, and tablets in a probability-based panel survey: Effects on survey measurement error,'' *Social Sci. Comput. Rev.*, vol. 34, no. 1, pp. 78–94, Feb. 2016.

[47] D. Shin, M. Choi, J. Hyun Kim, and J.-G. Lee, ''Interaction, engagement, and perceived interactivity in single-handed interaction,'' *Internet Res.*, vol. 26, no. 5, pp. 1134–1157, Oct. 2016.

[48] S. Sánchez-Gordón, M. Sánchez-Gordón, M. Yilmaz, and R. V. O'Connor, ''Integration of accessibility design patterns with the software implementation process of ISO/IEC 29110,'' *J. Softw., Evol. Process*, vol. 31, no. 1, p. e1987, Jan. 2019.

[49] T. Neil, *Mobile Design Pattern Gallery*, 2nd ed., Sebastopol, CA, USA: O'Reilly Media, Inc., 2015. [Online]. Available: https://www.safaribooksonline.com/library/view/mobile-design-pattern/9781449368586/

[50] J. Tidwell, *Designing Interfaces: Patterns for Effective Interaction Design*. Sebastopol, CA, USA: O'Reilly Media, 2020. [Online]. Available: https://books.google.com.pk/books?hl

[51] M. Donati, G. Mori, and F. Paternò, ''Understanding the transitions between web interfaces designed to stimulate specific emotions,'' *Universal Access Inf. Soc.*, vol. 19, no. 2, pp. 391–407, Jun. 2020.

[52] C. M. N. Faisal, M. Gonzalez-Rodriguez, D. Fernandez-Lanvin, and J. de Andres-Suarez, ''Web design attributes in building user trust, satisfaction, and loyalty for a high uncertainty avoidance culture,'' *IEEE Trans. Hum.-Mach. Syst.*, vol. 47, no. 6, pp. 847–859, Dec. 2017.

[53] N. Bonnardel, A. Piolat, and L. Le Bigot, ''The impact of colour on website appeal and users' cognitive processes,'' *Displays*, vol. 32, no. 2, pp. 69–80, Apr. 2011.

[54] C. M. N. Faisal, S. Shahid, J. D. A. Suarez, M. Gonzalez-Rodriguez, and D. F. Lanvin, ''The influence of mobile app design on emotions leads to purchase intention for risk avoidance cultures,'' *Int. J. Mobile Commun.*, vol. 24, no. 3, pp. 225–255, 2024.

[55] L. Punchoojit and N. Hongwarittorrn, ''Usability studies on mobile user interface design patterns: A systematic literature review,'' *Adv. Hum.-Comput. Interact.*, vol. 2017, pp. 1–22, Sep. 2017.

[56] S. McCarthy, W. Rowan, N. Kahma, L. Lynch, and T. P. Ertiö, ''Open e-learning platforms and the design-reality gap: An affordance theory perspective,'' *Inf. Technol. People*, vol. 35, no. 8, pp. 74–98, Dec. 2021.

[57] M. Benaida, ''Significance of culture toward the usability of web design and its relationship with satisfaction,'' *Universal Access Inf. Soc.*, vol. 21, no. 3, pp. 625–638, Aug. 2022.

[58] J. Nielsen, *Usability Engineering*. San Mateo, CA, USA: Morgan Kaufmann, 1994.

[59] A. Ospankhan and Z. Kalpeyeva, ''User preferences for UI design elements: Impact on learning and skill development on coursera, edx, and stepik,'' Satpayev Univ., Almaty, Kazakhstan, Tech. Rep., 2024. [Online]. Available: https://api.dspace.khadi.kharkov.ua/server/api/core/bitstreams/bb73540d-6d4a-471a-9256-227996ef4655/content

[60] M. E. Nenni, ''Integrating mental workload management in advanced human-machine interaction: The development process of the proof-of-concept to refine the concept,'' in *Proc. Int. Congr. Hum.-Comput. Interact., Optim. Robotic Appl. (HORA)*, May 2024, pp. 1–6.

[61] W. Ali, O. Riaz, S. Mumtaz, A. R. Khan, T. Saba, and S. A. Bahaj, ''Mobile application usability evaluation: A study based on demography,'' *IEEE Access*, vol. 10, pp. 41512–41524, 2022.

[62] S. A. G. Fatimah, A. Rahmati, and S. Bahri, ''Analysis of mental burden of students based on NASA-TLX at DAYAH DMA,'' *TEKNOSAINS, Jurnal Sains, Teknologi dan Informatika*, vol. 12, no. 1, pp. 46–53, Jan. 2025.

[63] M. Calvano, A. Curci, C. M. N. Faisal, A. Piccinno, and Q. Sohail, ''Usability aspects reduce design complexity and help prevent use-related errors in icu ventilators,'' in *Proc. 1st Int. Workshop Participatory Design End-User Develop.-Building Bridges*, 2024. [Online]. Available: https://ceur-ws.org/Vol-3778/short6.pdf

[64] D. Grant-Smith, T. Donnet, J. Macaulay, and R. Chapman, "Principles and practices for enhanced visual design in virtual learning environments: Do looks matter in student engagement?" in *Student-centered Virtual Learning Environments in Higher Education*. Hershey, PA, USA: IGI Global, 2019, pp. 103–133.

[65] J. D. Bader and P. R. Lowenthal, "Using visual design to improve the online learning experience: A synthesis of research on aesthetics," in *Visual Design for Online Learning*. USA: IGI Global, 2018, pp. 1–35. [Online]. Available: https://services.igi-global.com/resolvedoi/resolve.aspx?doi=10.4018/978-1-5225-4206-3.ch001

[66] A. Alshehri, M. Rutter, and S. Smith, "Assessing the relative importance of an E-learning system's usability design characteristics based on students' preferences," *Eur. J. Educ. Res.*, vol. 8, no. 3, pp. 839–855, 2019.

[67] L. F. da Silva, P. A. P. Junior, and A. P. Freire, "Mobile user interaction design patterns: A systematic mapping study," *Information*, vol. 13, no. 5, p. 236, May 2022.

[68] Y. Liu, H. Tan, G. Cao, and Y. Xu, "Enhancing user engagement through adaptive UI/UX design: A study on personalized mobile app interfaces," *World J. Innov. Modern Technol.*, vol. 7, no. 5, pp. 1–21, Sep. 2024.

[69] M. K. A. Braham, F. Buendía, and F. Gargouri, "Generation of adaptive mobile applications based on design patterns for user interfaces," *Proceedings*, vol. 31, no. 1, p. 19, 2019.

[70] *Design Patterns*. Accessed: May 25, 2025. [Online]. Available: https://ui-patterns.com/patterns

[71] C. Gurkan, "Development of Android-based Internet of Things application for data tracking in smart marble factories," *J. Artif. Intell. Data Sci.*, vol. 1, no. 1, pp. 41–44, 2021.

[72] N. K. S. Suleri, M. Jarke, and L. C. Tran, "UI design pattern-driven rapid prototyping for agile development of mobile applications," in *Proc. 21st Int. Conf. Hum.-Comput. Interact. Mobile Devices Services (MobileHCI)*, Jun. 2019.

[73] J.-E. Pelet, B. Taieb, M. Monia, B. D. M. Norchene, M. Prashant, Z. Zaid, and M. O. Joanna, "Impact of M-commerce website design on consumers' behavioral intentions: An empirical study of age as a moderating influence," in *Developments in Marketing Science: Proceedings of the Academy of Marketing Science*. Cham, Switzerland: Springer, 2017, pp. 111–124.

[74] S. Vecino, M. Gonzalez-Rodriguez, D. Fernandez-Lanvin, and J. de Andres, "The impact of serif vs sans-serif typefaces on e-commerce websites," *Int. J. Hum.-Comput. Interact.*, vol. 41, no. 5, pp. 3613–3624, 2025.

[75] M. Elias, K. Dick, and S. Paulo, "Influência das propriedades tipográficas na experiência do usuário em interfaces digitais," Tech. Rep., 2023, pp. 99–109. [Online]. Available: https://www.researchgate.net/publication/372680191_Influence_of_typographic_properties_on_user_experience_in_digital_interfaces

[76] J. Wang, J. Yin, S. Deng, Y. Li, C. Pu, Y. Tang, and Z. Luo, "Evaluating user satisfaction with typography designs via mining touch interaction data in mobile reading," in *Proc. CHI Conf. Hum. Factors Comput. Syst.*, Apr. 2018, pp. 1–12.

[77] R. P. Da Costa, E. D. Canedo, R. T. De Sousa, R. De Oliveira Albuquerque, and L. J. García Villalba, "Set of usability heuristics for quality assessment of mobile applications on smartphones," *IEEE Access*, vol. 7, pp. 116145–116161, 2019.

[78] M. L. Jensen, N. E. Dunbar, M. S. Connelly, W. D. Taylor, M. Hughes, B. Adame, and B. Rozzell, "Organizational balancing of website interactivity and control: An examination of ideological groups and the duality of goals," *Comput. Hum. Behav.*, vol. 38, pp. 43–54, Sep. 2014.

[79] A. Dirin, M. Nieminen, and T. H. Laine, "Feelings of being for mobile user experience design," *Int. J. Hum.-Comput. Interact.*, vol. 39, no. 20, pp. 4059–4079, Dec. 2023.

[80] A. M. Mutawa and S. Sruthi, "Enhancing human–computer interaction in online education: A machine learning approach to predicting student emotion and satisfaction," *Int. J. Hum.-Comput. Interact.*, vol. 40, no. 24, pp. 8827–8843, Dec. 2024.

[81] D. Minas, E. Theodosiou, K. Roumpas, and M. Xenos, "Adaptive real-time translation assistance through eye-tracking," *AI*, vol. 6, no. 1, p. 5, Jan. 2025.

[82] X. Zhan, Y. Xu, and Y. Liu, "Personalized UI layout generation using deep learning: An adaptive interface design approach for enhanced user experience," *J. Artif. Intell. Gen. Sci.*, vol. 6, no. 1, pp. 463–478, Dec. 2024.

[83] *The Impact of Interaction Design on Brand Perception*. Accessed: Jul. 26, 2025. [Online]. Available: https://www.nngroup.com/articles/interaction-branding/

[84] S. Fath, D. Abimanyu, and M. Misbak, "Exploring the relationship between responsiveness and usability and its impact on customer satisfaction in e-commerce," *J. Manage., Econ., Financial*, vol. 2, no. 4, pp. 72–79, Jul. 2024.

[85] E. G. Nilsson, "Design patterns for user interface for mobile applications," *Adv. Eng. Softw.*, vol. 40, no. 12, pp. 1318–1328, Dec. 2009.

[86] T. J. T. E. Rossiter and R. Fitzgerald, "Supporting university students' learning across time and space: A from-scratch, personalised and mobile-friendly approach," *Interact. Technol. Smart Educ.*, vol. 21, no. 1, pp. 108–130, Jan. 2024.

[87] S. Criollo, A. Guerrero-Arias, D. Buenaño Fernández, A. Jaramillo-Alcazar, and S. Luján-Mora, *Using Mixed Reality (MR) as an Emerging Technology for Improving Higher Education: Analysis of Mental Workload*. Ital Publication, 2024. [Online]. Available: https://rua.ua.es/server/api/core/bitstreams/aecae92a-9afd-4b28-813c-508f6b2544c7/content

[88] T. Alasmari, "The effect of screen size on students' cognitive load in mobile," *J. Educ., Teach. Learn.*, vol. 5, no. 2, pp. 280–295, 2020.

[89] A. S. Nimgaonkar and R. Kumbhar, "Usability—A key to security and utility: Case of core banking system," in *Proc. Int. Conf. Emerg. Smart Comput. Informat. (ESCI)*, Mar. 2024, pp. 1–6.

[90] O. Thinnukool and N. Kongchouy, "The User's satisfaction of graphic user interface in designing for health care mobile application," *J. Telecommun.*, vol. 9, pp. 11–15, Sep. 2017.

[91] B. A. Kumar and P. Mohite, "Usability of mobile learning applications: A systematic literature review," *J. Comput. Educ.*, vol. 5, no. 1, pp. 1–17, Mar. 2018.

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